Supporting Information for: Phenological sensitivity to climate change disrupts species' coexistence

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Empirical data

We obtained data germination data from (27). To estimate species level sensitivity and forecast phenological priority under alternative climate regimes (Fig. 1), we used a Bayesian mixed-effects accelerated failure time model modified from (28) using a Weibull distribution for the likelihood function. We included weeks of stratification and incubation temperature as fixed effects and species as a random effect. To estimate species-specific environmental responses, we let both the slopes and intercepts vary by species. See (7) for more details on model structure.

Model

For our example, we adapt the model from Wolkovich & Donahue 2020 (with more details therein), which is itself an adaptation of a common model of community dynamics, often conceptualized for annual plants with a seedbank.

We model a pair of competing species in which the seedbank N of species i at time $T+1$ (where T is the beginning of an annual growing season) is determined by the survival, s, of ungerminated seeds from the previous year, $1 - G_i(T)$, plus new biomass, B_i, produced from germinating seeds during the growing season (from T to $T + \delta$), which is converted to seeds at rate ϕ:

$$N_i(T+1) = sN_i(T)(1 - G_i(T)) + \phi B_i(T + \delta) \tag{1}$$

Within season dynamics of biomass are modeled as R^* competition that begins when resources become available at the beginning of the growing season, $R(t = 0) = R_0(T)$, where t indicates within-season time and $R_0(T)$ is initial resource availability in year T. For each species, biomass increases through germination $g_i(t)$ and resource uptake ($f(R)$), which is converted into biomass at a species-specific rate (c_i), less maintenance costs (m) (Figure S1b,c). Resource uptake per unit biomass is defined by the change in uptake per unit resource (a) and the maximum uptake ($1/u$).

$$\frac{\mathrm{d}B_i}{\mathrm{d}t} = (c_i f(R) - m)B_i + g_i(t) \tag{2}$$

$$f(R) = \frac{aR}{1 + auR} \tag{3}$$

The resource (R) itself declines across a growing season (Figure S1a) due to uptake by the standing biomass of both species and abiotic loss (ϵ):

$$\frac{\mathrm{d}R}{\mathrm{d}t} = -\sum_{i=1}^{2} f(R)B_i - \epsilon R \tag{4}$$

Our model departs from similar community models by explicit modeling of within-season germination phenology: germination, $g_i(t)$, is modeled as a time-dependent forcing function that adds daily biomass (germinants) according to a Hill equation, where the speed of germination (i.e., the day of half-germination, $\tau_{g50,i}$) is determined by a species-specific response to overwinter chilling.

$$g_i(t) = N_0(T)s_i b(g_{cum,i}(t) - g_{cum,i}(t-1)) \tag{5}$$

$$g_{cum,i}(t) = g_{max}(\frac{t^h}{\tau_{g50,i}(T)^h + t^h}) \tag{6}$$

where $N_0(T)$ is the number of overwintering seeds in year T, s_i is overwinter seed survival, b converts germinating seeds into biomass, and $g_{cum,i}(t) - g_{cum,i}(t-1)$ is the fraction of species i seeds germinating between day $t-1$ and t in the growing season (Figure S1e,f). The shape of the Hill Equation defining $g_{cum,i}(t)$ has the same maximum germination fraction (g_{max}) and Hill coefficient (h) for both species, and $\tau_{g50,i}(T)$ is the half-saturation constant for the Hill equation (i.e., the day when germination reaches half of g_max), and characterizes the species-specific delay in germination due to overwinter chilling in year T (Figure S2a,b). $\tau_{g50,i}(T)$ is Poisson-distributed and the expected value for germination delay, $\tau_{delay,i}$, declines with chilling, such that

$$\tau_{delay,i} = \tau_{max} e^{-\tau_{\xi,i}\xi(T)} \tag{7}$$

$$\tau_{g50,i} \sim \text{Poisson}(\tau_{delay,i}) \tag{8}$$

where $\xi(T)$ is the weeks of overwinter chilling in year T, $\tau_{\xi,i}$ is the species-specific sensitivity to chilling, and τ_{max} is the expected delay when overwinter chilling is zero. That is, germination depends on the interaction of a species trait and a varying environment, resulting in year to year variation in the speed of germination for each species (Figure S2e) and the timing of peak germination (Figure S2f).

Simulations:

To examine species coexistence when the timing of germination depends on annually varying cold stratification (similar to chilling), we ran 10000 simulations of two-species communities. Communities were assembled from randomly generated species, where each species varied in resource conversion efficiency (c_i) and in sensitivity to chilling ($\tau_{\xi,i}$) (see Table S1 for parameter values). Variation in resource conversion efficiency yields different R^* values, a metric of resource competition where a lower R^* means a species can draw the resource down to a lower level and is thus considered the superior competitor. Variation in sensitivity to chilling resulted in variation in the distribution of germination within a growing season (Figure S1d). All other parameters were identical between species (Table S1).

Within-year competitive dynamics were solved using an ode solver (`ode` in the R package `deSolve`) with initial condition $R = R_0(T)$ and $B_i(t = 0) = 0$, and germination was as a forcing function with germinants for each species added daily in proportion to a Hill Equation (Eqn 6, Figure S1d). The end-of-season biomass of each species was converted to seeds, added to surviving seeds that did not germinate in year T, and the populations were censused. At each census, a minimum cutoff was applied to define extinction from the model. Note that 'coexistence' in this model is defined by joint persistence through time and not by low density growth rate.

In all simulations, the environment varied from year to year in both resource availability, $R_0(T)$, and overwinter chilling, $\xi(T)$, such that

$$R_0(T) \sim logN(\mu_R, \sigma_R) \tag{9}$$

$$\xi(T) \sim logN(\mu_\xi, \sigma_\xi) \tag{10}$$

Variation in $R_0(T)$ and $\xi(T)$ were independent. We compared competitive coexistence in simulations from two different environmental regimes: a low-chill regime and a high-chill regime (Figure 2a, Table S1). Each species was initialized with a seedbank ($N_i(0) = 100$), and simulations were run for 300 years. For species that coexisted in the high-chill regime, we repeated the simulation of two species with the same resource conversion efficiency, c_i, and sensitivity to chilling, $\tau_{\xi,i}$, in a low-chill regime.

Figures

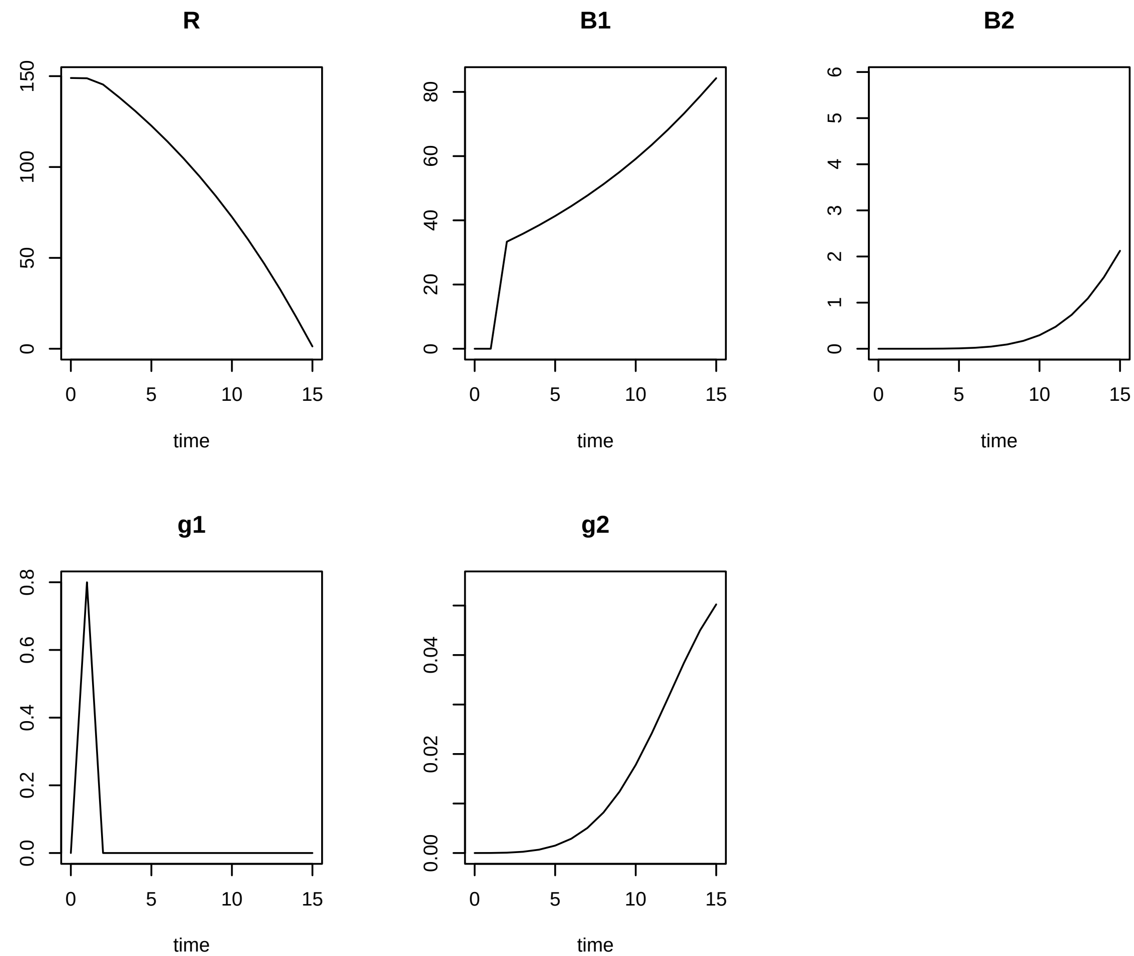

Figure S1: Within-season dynamics of (a) resources ($R(t)$, Eqn 3), (b,c) biomass of two competing species ($B_i(t)$, Eqn 2), and (d,e) the distribution of germination over the growing season for each species ($g_i(t)$, Eqn 6)

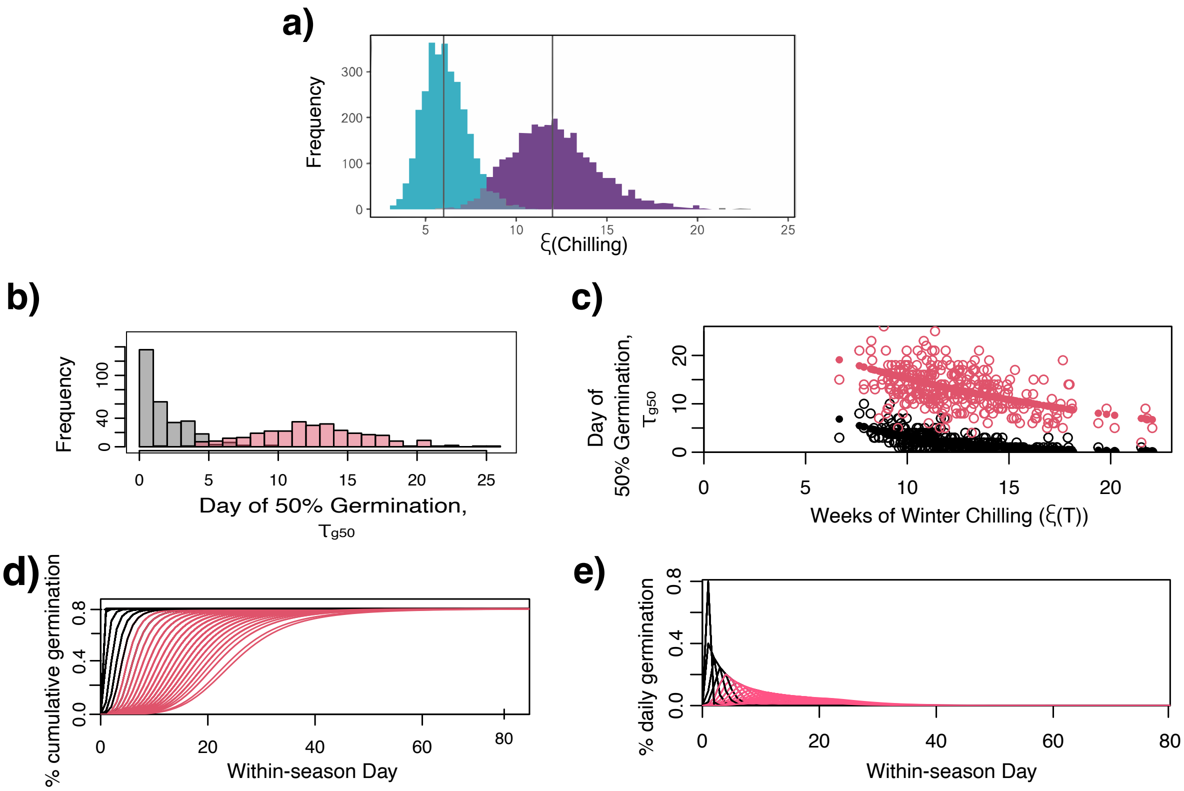

Figure S2: Environmental variation and species-specific responses. (a) Comparison of the distribution in winter chilling in low-chill (teal) and high-chill (purple) regimes. For a particular high-chill simulation run, panels (b) - (d) illustrate how two competing species (red vs black lines) respond to annual environmental variation: (b) Annual variation in the day of half-germination for two competing species; (c) Species-specific delay in germination in response to annual variation in the weeks of winter chilling; (d) Annual variation in cumulative germination over the growing season for two competing species; (e) Annual variation in daily germination over the growing season for two competing species.

1 Tables

Table S1: Parameter values, definitions, and units.

Parameter	Value(s)	Definition	Unit
T	1	annual timestep of population growth	year
t	1	daily timestep of growing season	days
δ	120	length of growing season	days
$N_i(T)$	$N_i(T=0) = 100$ $extinction threshold:$ $N_i(T) < 0.0001$	annual census of seedbank of species i	seeds per unit area
$G_i(T)$	g_{max}	seed germination fraction in year T	unitless
s	0.8	overwinter seedbank survival	unitless
$B_i(T+\delta)$	$\int_{t=T}^{t=T+\delta} \frac{dB_i}{dt}$	see Eqns 2,5	end of season biomass of species i from growth of germinated seeds, $G_i(T)$
ϕ	0.5	conversion of end-of-season biomass to seeds	biomass^{-1}
$R_0(T)$	$\sim logN(\mu_R, \sigma_R)$ $\mu_R = log(100), \sigma_R = log(2)$	amount of resource at the beginning of the growing season, varies annually	resource
ϵ	0.001	abiotic loss of R, Eqn 4	days^{-1}
c_i	$\sim$ Unif$(0.025, 0.75)$	conversion efficiency of R to biomass of species i	$\frac{\text{biomass}}{\text{resource}}$
m	0.005	maintenance costs during growth season for species i	days^{-1}
a	0.8	uptake increase as R increases for species i	days^{-1}
u	5	inverse of maximum uptake for species i	$\frac{\text{(days)(biomass)}}{\text{resource}}$
$\xi(T)$	$\sim logN(\mu_\xi, \sigma_\xi)$ $\mu_{\xi,lo} = log(6), \sigma_\xi = 0.2$ or $\mu_{\xi,hi} = log(12), \sigma_\xi = 0.2$	annually varying winter chilling	weeks of chilling
$g_i(t)$	Eqn 5	biomass of daily germinants of species i on day t of growing season	biomass
b	1	biomass of a germinant	$\frac{\text{biomass}}{\text{seeds}}$
$g_{cum,i}(t)$	Eqn 6	cumulative fraction of seeds germinated by day t	unitless
g_{max}	0.8	maximum germination fraction (max of Hill equation)	unitless
h	5	Hill constant for germination fraction	unitless
$\tau_{g50,i}$	$\sim$ Pois$(\tau_{delay,i})$	day of half-germination for species i, Eqn 8	days
$\tau_{delay,i}$	Eqn 7	expected germination delay from overwinter chilling for species i	days
$\tau_{\xi,i}$	$\sim$ Unif$(0, 1)$	sensitivity to chilling for species i	days/week